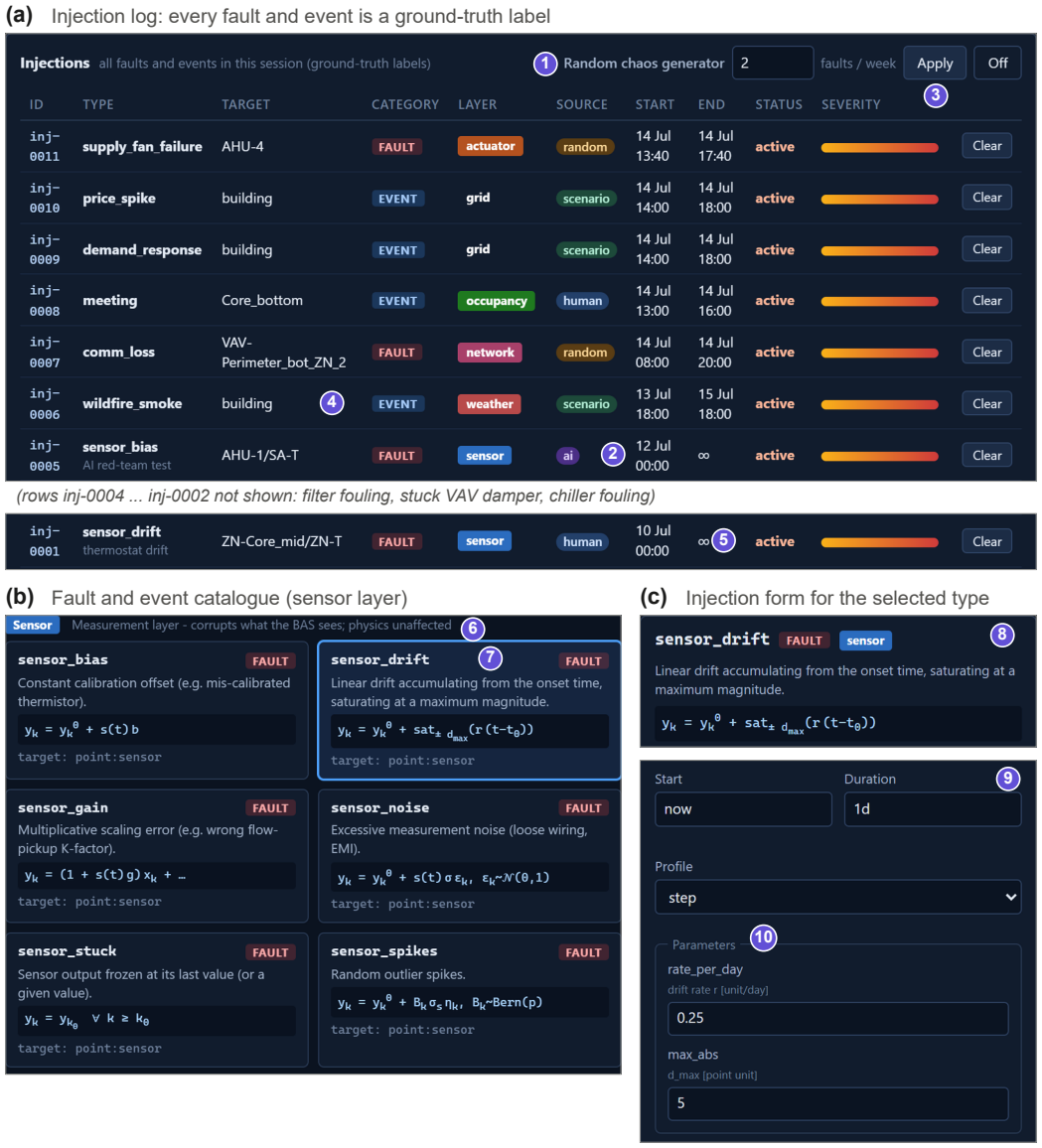

(a) Injection log: every fault and event is a ground-truth label
Injections all faults and events in this session (ground-truth labels)
1 Random chaos generator
2
faults / week
Apply
Off
3
ID
TYPE
TARGET
CATEGORY
LAYER
SOURCE
START
END
STATUS
SEVERITY
inj-0011
supply_fan_failure
AHU-4
FAULT
actuator
random
14 Jul 13:40
14 Jul 17:40
active
Clear
inj-0010
price_spike
building
EVENT
grid
scenario
14 Jul 14:00
14 Jul 18:00
active
Clear
inj-0009
demand_response
building
EVENT
grid
scenario
14 Jul 14:00
14 Jul 18:00
active
Clear
inj-0008
meeting
Core_bottom
EVENT
occupancy
human
14 Jul 13:00
14 Jul 16:00
active
Clear
inj-0007
comm_loss
VAV-Perimeter_bot_ZN_2
FAULT
network
random
14 Jul 08:00
14 Jul 20:00
active
Clear
inj-0006
wildfire_smoke
building
4
EVENT
weather
scenario
13 Jul 18:00
15 Jul 18:00
active
Clear
inj-0005
sensor_bias
AI red-team test
AHU-1/SA-T
FAULT
sensor
ai
2
12 Jul 00:00
∞
active
Clear
(rows inj-0004 ... inj-0002 not shown: filter fouling, stuck VAV damper, chiller fouling)
inj-0001
sensor_drift
thermostat drift
ZN-Core_mid/ZN-T
FAULT
sensor
human
10 Jul 00:00
∞
5
active
Clear
(b) Fault and event catalogue (sensor layer)
Sensor
Measurement layer - corrupts what the BAS sees; physics unaffected
6
sensor_bias
FAULT
Constant calibration offset (e.g. mis-calibrated thermistor).
$y_k = y_k^0 + s(t)\,b$
target: point:sensor
sensor_drift
7
FAULT
Linear drift accumulating from the onset time, saturating at a maximum magnitude.
$y_k = y_k^0 + \mathrm{sat}_{\pm\, d_{max}}(r\,(t - t_0))$
target: point:sensor
sensor_gain
FAULT
Multiplicative scaling error (e.g. wrong flow-pickup K-factor).
$y_k = (1 + s(t)\,g)\,x_k + \dots$
target: point:sensor
sensor_noise
FAULT
Excessive measurement noise (loose wiring, EMI).
$y_k = y_k^0 + s(t)\,\sigma\,\varepsilon_k,\ \varepsilon_k \sim \mathcal{N}(0,1)$
target: point:sensor
sensor_stuck
FAULT
Sensor output frozen at its last value (or a given value).
$y_k = y_{k_0} \quad \forall\ k \geq k_0$
target: point:sensor
sensor_spikes
FAULT
Random outlier spikes.
$y_k = y_k^0 + B_k\,\sigma_s\,\eta_k,\ B_k \sim \mathrm{Bern}(p)$
target: point:sensor
(c) Injection form for the selected type
sensor_drift
FAULT
sensor
8
Linear drift accumulating from the onset time, saturating at a maximum magnitude.
$y_k = y_k^0 + \mathrm{sat}_{\pm\, d_{max}}(r\,(t - t_0))$
Start
now
Duration
9
1d
Profile
step
Parameters
10
rate_per_day
drift rate r [unit/day]
0.25
max_abs
d_max [point unit]
5